

The module in ten weeks

WEEKS 1–3

Numerical foundations

- 1 Computation and simulation
- 2 Root-finding and fitting
- 3 Interpolation and integration

→ **Exercise 1 (25%)**

WEEKS 4–6

Uncertainty and hypothesis testing

- 4 Errors, tests, time series
- 5 Monte Carlo methods
- 6 Linear algebra and clustering

→ **Exercise 2 (25%)**

WEEKS 7–10

Inference and machine learning

- 7 Bayesian inference and MCMC
- 8 Classification
- 9 Method selection
- 10 Neural networks

→ **Exercise 3 (40%)**

Every week: lectures introduce the ideas, then a 3-hour lab in Python puts them to work on real data. Each numerical method is paired with the data-analysis question it answers.

ASSESSMENT

Coursework only: no exam

25%

Exercise 1

Weeks 1–3

Notebook + 300-word
interpretation

Due Fri 23 Oct, 4pm

25%

Exercise 2

Weeks 4–6

Notebook + 300-word
interpretation

Due Fri 13 Nov, 4pm

40%

Exercise 3

Weeks 7–10

Your own investigation: 5-page
report + code

Due Mon 11 Jan, 12pm

10%

In class

Weeks 6–7 and 8–9
(5% each)

5–8 minute discussion of your
work, based on your logbook

Start your logbook today



A feedback session follows each exercise deadline: Fridays at 5pm.



GenAI: fine for help with code. Your interpretation must be your own. Declaration with every submission.

TODAY'S LAB

Getting started

- 1 Open the module repository and start with **week1_setup.ipynb**
- 2 **File** → **Save a copy in Drive** before you change anything
- 3 Make your copy of the **logbook** (setup notebook, Section 6)
- 4 Then: tidal data → revision → break → foundations → Euler vs RK4



[github.com/MatSmithAstro/
phys7101_2026_resources](https://github.com/MatSmithAstro/phys7101_2026_resources)

Two PGTAs are here for the whole session: ask early. Nothing today is assessed.